

The Card Services Platform — rapid delivery of giant systems

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This paper describes the top-level integration strategy for delivering the Card Services Platform (CSP), formerly known as Cashless Services Replacement System (CSRS), probably the largest successful integration by a single BT team to date. This paper explains the need for such a strategy to evolve, leading to significant efficiency improvements. It identifies the need for the careful management of integration testing of the complex interfaces involved in a system comprised of over 40 large computer systems, spread across the country. It shows how potential operational difficulties can be identified in the system testing phase and eradicated before impacting the product in a 'live' situation. It outlines testing techniques, test coverage, granularity of test results and 'measure of goodness' of the tested product.

1. Introduction

What has 6 square inches of plastic, 12 million holders, 40 separate systems, 100 interfaces, multiple operating systems and is dispersed across sites up and down the country? The BT Chargecard 'super-system', more formally known as the Card Services Platform (CSP). This paper tells the story of card services integration from early design up to the present day.

In early 1994, a small core team of BT integration engineers from BT Laboratories went to spend several weeks at MCI's development laboratories at Cedar Rapids, Iowa, with the aim of understanding how MCI built and ran their intelligent services network (ISN), the heart of their card services product.

By the summer of 1994 the first full end-to-end design for a BT integrated platform had emerged from the Belfast-based card services design team. The plan was to:

- reuse as much of the BT operational support systems (OSS) architecture as possible,
- reuse as much of the MCI ISN software as could be managed, within the limitations of a UK dialling plan and network,
- manage the customer base on BT's core operational support system — customer service system (CSS).

Development work began in MCI, Nortel (NT), Ericsson (ETL), and BT (in Belfast, Cardiff and London) during July-August 1994, with a large systems integration team being pulled together to manage the build, integration, and test of the 30-40 systems scheduled for delivery in autumn 1994.

The main systems integration team consisted of around 60 full time engineers from BT's Systems Integration department, with more in part-time support from the Application Support Group (ASG), Computing Services and Operations (CSO), and Networks and Systems Operations (N&SO). The working approach was documented [1], being based firmly on the principles of the Systems Integration Handbook [2]. This provided for functional partitioning of the system into the following sub-systems:

- network elements,
- network management,
- billing and service management.

These functions were backed up by the key orthogonal role of systems build.

The plan was to integrate and test all connected components (i.e. single computer systems, e.g. CSS) within a sub-system, to a strict timeline. At the end of this, following satisfaction of a quality gate specification, the separated sub-systems would be connected together to form the complete Chargecard 'super system'. A distinct system test team would then execute a set of 'end-to-end' tests: beginning with order placement, through telephone call origination and completion, to bill production, backed up by management information statistics and network management monitoring. The systems build team kept the sub-systems and later the whole system, hanging together and working consistently over all their networks, both remote and local. A key part of this role was managing the

interface and support from the Network Integration Facility, CSO and the ASG teams.

The challenge was huge and probably unprecedented in BT as a single integration project. Time and planning were of the essence, the aim being to deliver a fully working system into the UK market before spring 1996, just in time to replace the ageing and limited Chargecard Services Database (CSDB) system. Apart from the planned rapid expansion of the UK card base, a more flexible platform was required to ensure BT was ready to compete in a card services market that was about to become highly competitive with potential interest from large operators such as AT&T.

Delivery of most of the major systems from the development teams started in earnest around September/October 1994, though delays and some late requirements kept this flow extended over the period up to April 1995.

Test preparation was a big effort for such a large and complex system. Even when broken down into sub-systems, many tests were still being 'tidied up' when sub-system testing began in Autumn 1994. Full end-to-end system testing was completed on schedule in July 1995. Cut-over, from the test environment to the 'live' environment, was a complex and lengthy task; the system being spread over so many sites. Cut-over began in early August. The system was signed off for 'live' customer trial (6000 Chargecards, rising to 14 000 at end of trial) on 12 September 1995. Parallel running of CSP alongside the old CSDB system lasted for five months. Full service launch and migration of the remaining 6000 000 customers to CSRS occurred on 5 February 1996. This was rapidly followed up by a huge marketing launch for the new Chargecards that has increased the card base to around twice that number today.

The complexity of moving existing services on to the new architecture, and adding to these the new services required to be supported by the card services platform (CSP), was such that it was necessary to take a phased approach to their implementation and subsequent integration and testing. These phases were identified as 'Day 1' and 'Day 2'.

2. CSP Day 1 — feature set

For CSP Day 1, it was intended that the platform, with its capability of supporting much larger volumes of traffic and more features than CSDB offered, should support as many of the existing BT Chargecard features as possible. Not all existing features could be migrated to the platform for Day 1. The ones not possible to migrate in time (e.g. Visa phone, credit call, and the 'adult' opt-in services) were to remain operational on CSDB. They would transfer to the new CSP platform for Day 2, along with the new features.

Three types of CSP card were to exist for Day 1:

- contact card,
- national card,
- full service (international) card.

These replaced the six types of BT Chargecard supported on the CSDB system. The BT Chargecard allows calls to be made using the CSP; these are calls originating in the UK or in countries supporting the existing 'BT direct' service. The original BT direct service allowed callers abroad to access a BT international operator, using a toll-free access number. For Day 1, one of the new services introduced was 'BT Auto Direct'; this allowed callers to dial directly into the CSP platform for automatic call processing, thus saving on operator expense.

The intelligent network functionality, delivering the advances in CSP user functionality and flexibility, is provided by the MCI intelligent services network (ISN). This formed the bulk of the network element sub-system.

The ISN specific features supplied for Day 1 included:

- language selection — such as Welsh, French,
- re-origination — dialling '##' informs the ISN that a new call to the ISN is requested. This saves the user having to re-enter their card number and PIN,
- speed-dialling — utilising 9 short codes ('*91' to '*99') to hold full-length telephone numbers,
- drop back to operator — if card number and PIN not valid or if the telephone does not support tone dialling,
- on-demand operator assistance — by entering # before entering card number and PIN, or *0 after card number and PIN entered.

3. CSP Day 1 — logical architecture

Figure 1 shows the complete logical CSP system architecture. Essentially the system consists of a public switched telephone network (PSTN) interface to feed the ISN call-handling platform and operator services terminals. All switch components involved in this process are monitored using BT standard network management tools.

The ISN feeds fraud detection data alongside its call record link with the billing systems. Service management provides card details and associated updates to ensure appropriate card validation.

The management information system (MIS) collects card and call data to act as a basis for marketing decision making.

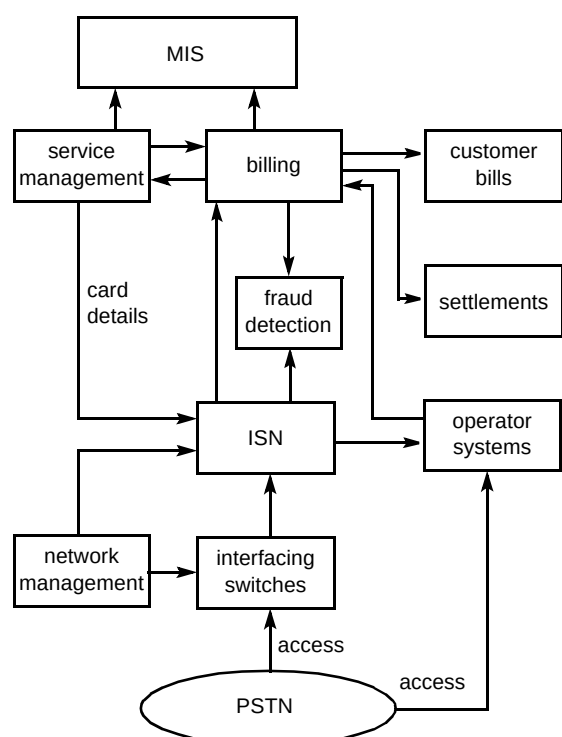


Fig 1 Logical model.

Settlement systems provide inter-country call volumes so that international bulk traffic costs can be appropriately charged.

The architecture for Day 1 and the services supported by it are defined in detail in the end-to-end design document [3].

4. CSP Day 1—physical architecture

Card details are entered via CSS. Multiples of cards may be allocated against each billable phone number. The details are then fed from CSS to the ISN, and machines in the billing stream. Figure 2 gives a much-simplified overview of the physical stages in handling a CSP call.

The physical architecture is best explained by describing the flow of a typical call.

The PSTN delivers a national or international incoming call to the ISN platform via the signalling converter (an ETL AXE10 switch). This maps associated message fields between the BT (C7) and USA (SS7) signalling protocols. The bridge switch (an NT DMS250 switch) forwards the call to the call distributor (an NT DMS100 switch) which chooses the correct destination group of ports for the call type in question. Most calls are passed to the automatic call-handling part of the ISN to validate the card number/PIN and destination number as well as check for possible fraud.

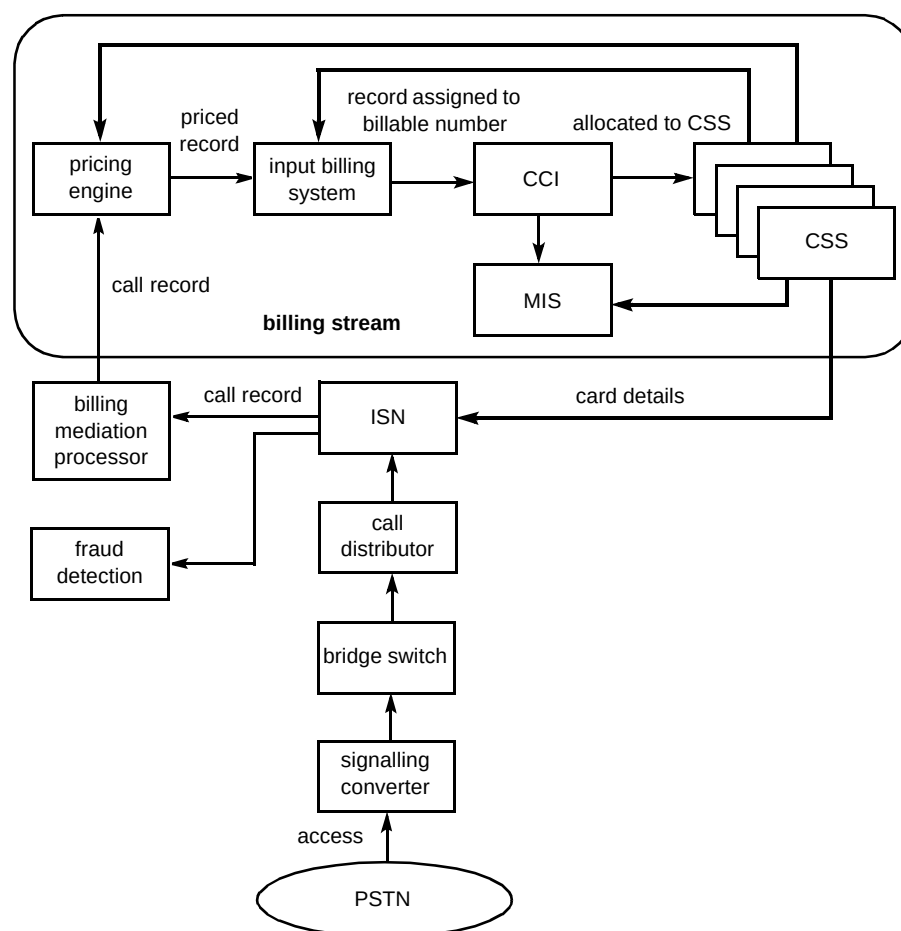


Fig 2 Physical architecture.

A new call is then set up through the bridge switch to the intended PSTN destination and the ISN port and associated call distributor circuits released. Once answered this new call is bridged with the original incoming call so that speech can proceed. Re-origination is now possible by detection of ‘##’ at the Bridge switch, disconnection of the destination half of the call and re-forwarding of the original incoming call to the ISN platform.

On call completion call records from the ISN and the bridge switch are merged and passed to the billing mediation processor for conversion into a BT standard billing record format. Such records can then be priced by the pricing engine and matched to their associated billable telephone number by the input billing system. In most cases this resides on the main BT billing system within the customer service system (CSS) where the call record details are added ready for billing to the customer.

5. CSP Day 2 — enhancements

The scope of Day 2 was so large that it was split into phases, namely, 2.0, and 2.1. Both phases were released in a progressive manner. Day 2 feature enhancements have already added the following:

- potential for ‘store-and-forward messages’ to be left on a speech application platform [4] if the call destination is unavailable for whatever reason,
- the ‘Opt-in’ product and its attendant card migration,
- ‘in-language’ prompts based on access details to support the European calling card product,
- national-only contact card functionality to restrict potential call charges for a new ‘Ring Me Free’ card product.

Note that the last two items have simply involved reconfiguration rather than new code delivery.

6. Integration test approach

6.1 Test environment and test tools

For Day 1, the CSP testers did not have the luxury of a self-contained complete test model, as was achieved for Day 2. The Day 1 tests were performed on one of the two ‘live’ production ISN platforms and associated switch stacks. This had the advantage of giving the operational teams experience in managing the intended ‘live’ environment equipment and allowed for limited but representative call volume tests. However, the billing, service management and network management sub-system test environments were self-contained even if distributed throughout the UK.

The Day 2 environment included a scaled down ISN platform at BT Laboratories connected to the original environment for the other sub-systems, although this had to

be thoroughly regression tested to ensure it matched the known ‘live’ characteristics.

A third environment combination was employed by the network elements sub-system team using the BT Signalling Converter interworking with MCI’s ISN test environment in USA. This enabled early detection of significant problems prior to code delivery in the UK as well as a dual test route for new releases while the BT ISN was in use for current release verification.

Test automation was used where easily applicable, such as in the switch stack testing. One significant aspect of ‘system test’ was to ensure that the system could handle the expected level of call traffic. A call sender, using software-driven call types, generated in the region of 20 000 successful call attempts per hour via the AXE10s. This was less than the amount ideally required and revealed in good time that the switches had a performance aspect that needed to be resolved.

6.2 Design of the tests

Sub-system tests focused on detailed verification of the functional features of the components involved and their clean interworking. Subsystem-to-subsystem interfaces were also included, where possible prior to system test. In particular, there was a small range of cross-subsystem tests defined in the quality gate specification. System test of the fully integrated system was centred on test cases with a one-to-one mapping to features declared in the product definition [5, 6]. This provided full requirement traceability.

This mapping enables test design to be carried out for core areas of functionality and this was enhanced by the use of standard test variable lists, generic procedures and test matrices as described below.

Requirements traceability

Being able to provide adequate traceability back to requirements [5, 6], for the testing carried out, is important as it ensures that:

- all requirements are tested,
- no unnecessary testing is being carried out.

Requirement traceability is documented by means of a requirement matrix hierarchy, as illustrated in Fig 3a, 3b and 3c. At the highest level, a matrix is produced which maps requirements uniquely numbered in the product definition [5, 6] e.g. AD-CS-700, to system test designs, e.g. ‘CSS create card’. Similarly, in Fig 3a, requirement AD-CS-703 is mapped to domestic call completion. This mapping is indicated by the shaded box.

	domestic call completion	international incoming call completion	international outgoing call completion	3rd country international call completion	call transfer to MTOC	follow-on call completion	change speeddial nos, change language	CSS create card
AD-CS-700								
AD-CS-701								
AD-CS-702								
AD-CS-703								
AD-CS-704								
AD-CS-705								
AD-CS-706								
AD-CS-707								
AD-CS-708								
...								

Fig 3a Requirements matrix hierarchy — 1.

domestic call completion - requirements matrix					
test case	AD-CS-00703	AD-CS-00706	AD-CS-00707	AD-CS-00708	AD-CS-00709
DOM.ECC.1					
DOM.ECC.2					
DOM.ECC.3					
DOM.ECC.4					
DOM.ECC.5					
...					

Fig 3b Requirements matrix hierarchy — 2.

CSS create card - requirements matrix				
test case	AD-CS-00700	AD-CS-00701	AD-CS-00715	...
CREATE.ECC.1				
CREATE.ECC.2				
...				

Fig 3c Requirements matrix hierarchy — 3.

Thereafter, each owner of a system test design, e.g. 'CSS create card', is responsible for producing a matrix, for that test design, which maps test case identifiers, e.g. CREATE.ECC.1, CREATE.ECC.2, to applicable requirements, e.g. AD-CS-00701. It is often the case that a re-requirement will need to be tested as part of more than one test design.

Test designs

All aspects of the system needed to be tested, e.g. card creation/deletion, automatic access to the ISN, operator services, fraudulent call detection, correct pricing, and bill production.

Additionally, the system had to guarantee an exceptional level of integrity in terms of not losing any call records. This is achieved using an audit tool capturing details of call records passing along the billing path, and an error-handling tool that receives records that a billing component could not handle at that moment.

The system test design overview [7] summarised all the call types that were to be made for a range of customer types. It was essential to verify a wide range of call types (e.g. a 'bill to credit card' type customer making a call from a UK-managed payphone to a Dutch telephone, via an international operator). Clearly the number of combinations are massive. Hence, care had to be exercised in choosing the value of each of the relevant variables for each test case. The approach taken was to ensure that all possible values of variables that affect a particular system test design are tested at least once.

Twenty system test designs were identified, each one focusing on a core area of CSP functionality. One such test design sample was the 'domestic call completion billing and settlement test design'. This test design contains tests proving that the Card Services Platform supports automatic (non-operator-completed calls), domestic calls (i.e. calls originating from a UK customer premises equipment (CPE), and terminating on a UK CPE), and that the billing and settlement are made via the appropriate paths.

The scope of each system test design was defined so that:

- duplication of testing across test streams was minimised,
- all areas of testing were covered, with no gaps.

To ensure a common approach to test specification a 'CSP Integration Master Variable List' and a set of generic test procedures were created. This enabled the construction of a test summary in the form of a matrix.

The CSP Integration Master Variable List

The test variables list contains over fifty entries ($t_1, t_2, t_3, \dots$) which identify the conditions which can be changed for each test, e.g. t_4 = Chargecard type, t_{20} = Language Selected. Each test variable has a set of possible values ($v_1, v_2, v_3, \dots$) (e.g. the values that the Chargecard type can have include v_1 = National, v_2 = International, v_3 = Contact, v_4 = Opt-in).

The test matrix

For each system test design, there is a set of test cases. These test cases are contained in a matrix (see Fig 4), where each row represents a test case and each column represents a test variable such as Chargecard type. The number of columns is dependent on the particular test design being specified and only a subset of the total number of variables is usually required per test design.

The test matrix, together with the Master Variables List allows for rapid test specification.

Generic test procedures

The generic test procedures were devised to eliminate the unnecessary duplication of work. They are generic because they can be used by many of the test cases. These procedures take account of the values of variables being used in a particular test, e.g. one row in a test matrix, and, therefore, often take on the appearance of pseudo-code.

Example: procedure to make an International call

Step 1 Dial the **access number** (t_1)

If **ChargeCard** (t_2) = **International** (v_1) then

Hear announcement 1

If **ChargeCard** (t_2) = **National** (v_2) then

Hear announcement 2

If **ChargeCard** (t_2) = **Contact** (v_3) then

Hear announcement 3

6.3 CSS acceptance testing

The CSS programme supports functionality for a wide range of products. Each of these must be regression tested, prior to the rolling out of a new release. The CSP integration team performs the regression of Chargecard functionality for each new release. In the same time window, other test teams undertake 'user acceptance testing' (UAT). All tests must be completed before CSS can be signed off, for release to the 'live' environment.

6.4 Feature interaction testing

Although the prime focus for the integration of CSP is Chargecard internal functionality, there is a need to verify expected interworking with other, existing, and PSTN products (e.g. CallMinder). Items such as this will be contracted out to a specialist test group, in order to make use of existing skills and thus save test preparation effort.

6.5 Test documentation

The documentation for CSP system test was hierarchical in that key documents outlined the general requirements: the product definition [5, 6], the overall strategy for VV&T [1] and the CSP system test overview [7]. The detailed tests were in individual test design documents specifically focusing on low-level detail of that test — what the test was for, how the procedure was to be carried out, and the expected results for a successful test case. Test result documents were produced for versions of components tested. The documents stated not just the version of the

System test design — matrix											
	t_1	t_2	t_3	t_7	t_8	t_{10}	t_{12}	t_{14}	t_{15}	t_{25}	t_{26}
Test. 1	V_1	V_1	V_3	V_3	V_1	V_4	V_3	V_3	V_1	V_2	V_4
Test.2	V_2	V_1	V_3	V_3	V_1	V_3	V_1	V_1	V_3	V_2	V_3
Test.3	V_3	V_2	V_3	V_3	V_1	V_2	V_1	V_4	V_3	V_3	V_1
Test.4	V_4	V_2	V_3	V_3	V_2	V_4	V_4	V_3	V_3	V_4	V_4
Test.5	V_5	V_2	V_4	V_4	V_3	V_1	V_4	V_3	V_3	V_3	V_2
Test.6	V_3	V_1	V_1	V_2	V_2	V_4	V_2	V_3	V_4	V_2	V_4
Test.7	V_3	V_1	V_4	V_2	V_3	V_3	V_3	V_3	V_4	V_3	V_1
Test.8	V_4	V_1	V_3	V_3	V_1	V_2	V_4	V_4	V_3	V_3	V_1
Test.9	V_4	V_3	V_3	V_3	V_4	V_3	V_3	V_2	V_3	V_2	V_2

Fig 4 Test case matrix.

component tested and its results, but also included a compatibility matrix indicating the versions of interfacing components with which it had been tested. This enhanced the prospects of upgrades to 'live' environment machines going smoothly.

Each full end-to-end release of the whole CSP system is fully documented, in the release statement [8] produced for that release. These contain a full description of end-to-end test coverage, component versions tested, reference model limitations, and outstanding problems.

6.6 Configuration management and problem reporting

For a programme the size of CSP, it is especially important that good configuration management (CM) practices are in place. The CSP programme used the configuration management unit (CMU) at BT Laboratories. This ensured standard CM was put in place for all CSP-related teams — development, test environment installers, and testers.

Problem reporting was also based around CMU tools, namely the PCMS product. This provided programme-wide access to, and tracking of, problem reports. It also enabled management reports and statistics to be easily obtained.

7. Schedules and metrics

This section describes the planned and actual schedules that were followed for the Day 1 and Day 2 integration activities. It elaborates the fault profile obtained and the conclusions that were drawn from them to refine the test approach.

7.1 The schedule for Day 1

In practice, the sequential approach depicted in Fig 5 and described earlier was not fully implemented.

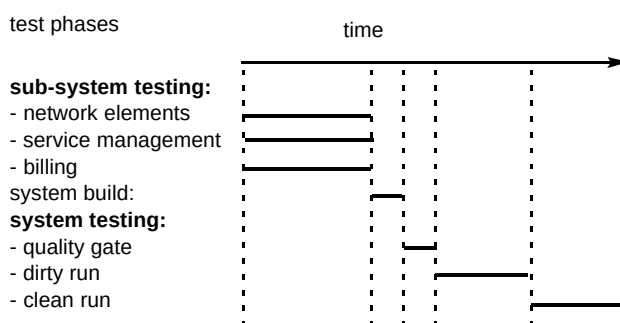


Fig 5 Integration master schedule (planned).

Instead, some of the test phases ended up being run in parallel with the system test phase, as depicted in Fig 6. As

but the decision was taken to continue with the system test phase.

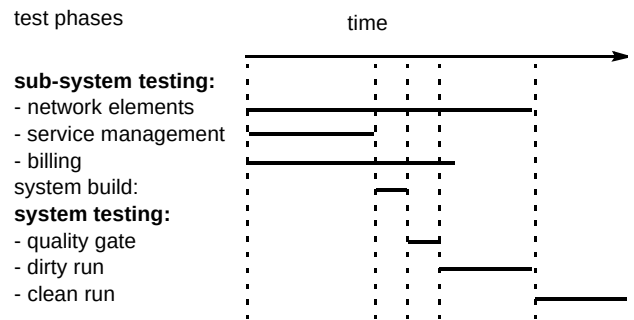


Fig 6 Integration master schedule (actual).

A consequence of this was that major problems were not being found in the early stages of sub-system testing as expected. Instead, problems were being found during the end-to-end system testing. The number of problems found continued to increase even in the latter stages, in the so called 'clean run', resulting in a problem report (PR) profile which clearly did not map to that expected. A comparison of the actual PR profile and the expected PR profile is illustrated in Fig 7.

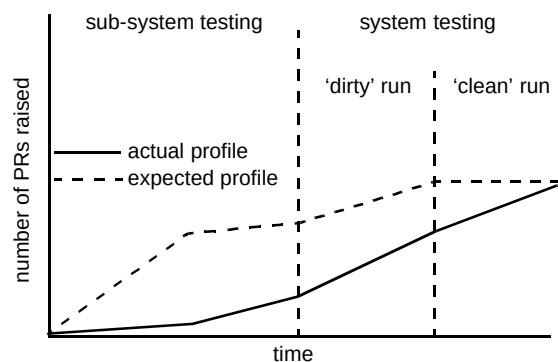


Fig 7 Comparison of problem report profiles.

Fault metrics at the end of the sub-system test showed around 400 significant faults (in total) in all of the sub-systems. This number was to rise to 1000 faults by the end of the system test.

The knock-on effect of this was that the planned dates for service launch were in jeopardy, due to the number of outstanding major problems at the end of the whole integration and test phase.

Despite the critical stage of testing, teamwork and close relationships with operations and the product line enabled tactical decisions to be made (e.g. early release of software to operational readiness testing (ORT), followed by a stable release still within ORT, which enabled the release to go out on its planned date).

7.2 Learning points from Day 1

There are seven key areas where experience has led to changes in the strategy.

- The timing of major problem discovery showed clearly that there was a need to run end-to-end system tests as early in the testing life cycle as possible. Hence, it was decided that new software deliveries would only have a sub-system sanity check on their arrival into integration, to enable end-to-end test streams to start as quickly as possible. However, this is only applicable when a full system has already been delivered, and subsequent releases are 'deltas' to a known baseline.
- The involvement of network operations, application support groups, operators and customer service agents in the test design creation and physical testing/platform management provided them with advance product information and also vital training opportunities. These included the necessary system exposure to improve the quality of training for new personnel and associated operational procedures.
- Automated regression testing procedures enabled new component releases to be rapidly verified before their inclusion in the system test. These procedures were also an essential part of the verification of maintenance releases of components after Day 1.
- The need to get to grips with many unknowns in the important technical detail — organising the operational schedules for the batch processing, set-up of co-ordinated configuration data on the many (big) systems, where to probe the system for best test data analysis during the partial integration stages, etc.
- Engineering decisions need to be taken to determine what should be the scope of a 'reasonable' test environment. For example, should we test out to all settlement systems? Should we scale the components close to 'live' size, and if we do not, how confident do we feel about performance? Do we need multiples of CSS images (expensive — but, there are 29 in 'live')? Should we use CA7 scheduling to auto-schedule our mainframe programs as in 'live' (or will this inhibit testing)? Should we treat job control language (JCL) as code, and test it as part of sign-off for 'live'?
- The developed process of how we were to interwork with the hugely dominant CSS programme, given that we would often need change in each CSS release, and the CSS release cycles are shorter in time-scale than a CSP release — CSS does not have a Chargecard test environment, and yet it needs to ensure that changes for CSP in CSS are tested and signed off before each CSS pilot.

- The need to understand all of the peripheral, but important 'extra' criteria which need to be taken into account before a release of CSP could be approved for BT network use — such items as audit integrity for BT's 'watchdog', finance billing and receivables (FBR), and the required tie-up of new card releases with the appropriate tariff release in BT's national charging data base (NCDB) had to be co-ordinated with each CSP release.

It is true to say that a lack of attention to fine detail from the start could easily have put large areas of the work at risk. This would have almost certainly caused severe slippage to the delivery schedule.

7.3 The schedule for Day 2 (R2.0)

The R2.0 part of Day 2 provided for store-and-forward messaging and the Opt-in product. The integration strategy was essentially unaltered. A sequential approach to testing remained, with the most significant change being that the time frames for each phase of integration and testing were shortened. Figure 8 makes the comparison between Day 1 and Day 2 (R2.0) time frames (in weeks).

Integration phase	Day 1	Day 2(R2.0)
Sub-system testing	20	10
System testing	12	6

Fig 8 Day1/Day2 time frames.

It was thus essential to:

- find problems earlier in the test life cycle,
- reduce the amount of testing, by being intelligently selective in what testing is run,
- reduce/remove duplication of testing by the sub-system test, system test and customer acceptance testing teams.

The tighter time-scales were achieved because of the identification of the learning points from Day 1. In addition, the system no longer required 'building' in the true sense of the word as it already existed in a fully functional Day 1 state. New software releases were mainly normal upgrades to the day 1 versions. However, the introduction of the speech applications processor [4], for store-and-forward messaging, required a new box to be added to the environment, and the integration of two new interfaces.

7.4 The schedule for Day 2 (post-R2.0)

After Release 2.0, the strategy was improved still further, by adopting a faster and more efficient route to

system test for the European calling card and Ring-Me-Free products. This involved adopting a ‘call stream’ approach whereby sub-system testing was reduced considerably, and, in many cases, appeared as shortened system tests. Sub-system teams ‘owned’ the system test call-streams applicable to their own specialist area of knowledge, and made test calls, etc, as soon as enough of the system has been upgraded. It is under their control, i.e. not under the control of a separate ‘system test team’. This approach has:

- shortened time-scales still further (though relatively less than the change from Day 1 to R2.0),
- enabled big faults (end-to-end type faults) to be found earlier,
- enabled VV&T team leaders to ‘own’ end-to-end test specification, and results reporting, for specific streams of customer product functionality pertinent to their special knowledge, e.g. the call barring system test stream was owned by network elements VV&T team leader,
- enabled more efficient use of teams,
- spread better ‘end-to-end’ knowledge across all teams.

7.5 *Proposed additional future schedule improvements*

The existing CSP integration team continues to make small efficiencies almost every month in the programme, but the overall approach is now clearly set in a mould that delivers a quality system to an acceptable time frame.

The programme and especially the customer-facing divisions continue to look for a much more rapid delivery of new products from test into ‘live’. Nearly three years on, with four major releases and nine minor releases completed, a large part of the work is regression testing of existing functionality. Sometimes, if the work is due to a major hardware upgrade, then all the testing is regression testing. Test automation, for this testing especially, is being steadily enhanced in the production of test calls, entry of customer data, verification of voice scripts, and ultimately the automatic checking of manual terminal text scripts. Automating billing record verification is then the next logical step. This approach is significantly reducing the manual test execution effort required. With increasing compatibility complexity, this regression aspect is becoming more and more important.

An obvious indirect test saving can be made when multiple new products are included within a single integration and test run. This incurs just one set of regression tests, pricing runs, audit integrity tests and ‘live’ sign-off and cut-over. There is a careful balance required, however, because incorporating too much functionality into a single release causes:

- complexity increase — non-functionally cohesive products are likely to add considerably more test variables to those that need testing, possibly increasing the number of test-cases by a large number,
- an increase to development delivery time-scales, which are likely to get out of synchronisation with each other, precluding a reasonably ‘clean start’ to the integration activity — a grossly staggered start to integration can actually extend time-scales,
- more to go wrong — a priority-1 fault in one product can cause serious delays to the delivery of other products,
- confused requirements — the product definition becomes large and difficult to track, with the number of requests for change (RFCs) against the release correspondingly increasing,
- more opportunity for the ‘ripple effect’ of faults in one product interfering with, or delaying the introduction of, another product.

8. **Cut-over strategies**

The Day 1 product was cut over between two disparate systems with little commonality, and with no ‘live’ traffic integrity to be maintained during launch. This removed most issues of compatibility verification.

The major difference between performing cut-over for Day 1 and performing it for Day 2 (R2.0) was that for Day 2 the Card Services Platform was now ‘live’ — managing ‘live’ customers, ‘live’ traffic and ‘live’ data. This difference became a real issue when the strategy for cut-over, i.e. the upgrade of components in a live environment to invoke the new functionality, was being developed.

The cut-over strategy worked out with the operational units affected, aimed to stagger the upgrade of systems in the live environment as far as possible. This would reduce risk and ease the management of the upgrade. However, the staggered approach entailed having both old and new systems working together for periods of time until all of the new systems were upgraded. This approach raises the common hardware/software issue of ‘backward compatibility’.

For R2.0, the integration team had successfully tested that all the upgraded components worked together in a test environment, but they had not necessarily tested that if component *A* had an interface with component *B* that the new component *A* worked with old component *B* or vice versa. If a staggered roll-out was to be achieved, a defined list of compatibilities must be proved. The ISN and CSS components were upgraded according to a staggered roll-out schedule. Due to the complexity of the compatibility issues for achieving a staggered roll-out of the billing

systems, they had to be upgraded as a unit, i.e. 'big bang' approach, with the attendant risk that introduced.

R2.0 was quickly followed by the European Calling Card (ECC) product. For ECC, a staggered approach to cut-over was successfully achieved. This was by virtue of improved communication between ASG, the cut-over manager, and test manager. This led to an agreed order of upgrades in the 'live' environment being mirrored in the test environment, with the consequent combinations of equipment and their compatibility issues.

ECC required only configuration upgrades on the ISN, but code upgrades were required for the CSS and Billing systems. The same is true for the Ring-Me-Free product.

The major upgrade as part of Day 2 — termed R2.1 — will require significant ISN staged cut-over planning and this is now at an advanced stage. The billing systems will probably be upgraded as a unit again, but CSS will roll according to its normal company-wide schedule.

9. Conclusions

Despite the size, complexity and challenging schedule set for the integration of the CSP system, the adoption of an ordered best practice approach has paid off in terms of repeated delivery success to aggressive time-scales. The use of established VV&T processes from the start enabled the team to understand where the scope for efficiencies existed in the releases following Release 1. These lay especially in the areas of:

- finding problems earlier in the test life cycle,
- reducing the amount of testing, by being intelligently selective of what testing is undertaken,
- reducing/removing duplication of testing by the sub-system test, system test and customer acceptance testing teams.

Careful assessment of the balance between new functionality and increasing product interdependency is critical for each new release. The difficulty is getting the right mix of efficiency saving and complexity. With CSP it has largely become a top-team consensus decision based on 'live' system needs versus an understanding of what can be delivered in a reasonable time frame by the key development communities. For the most part this has worked out well in practice.

With the programme and especially the customer-facing divisions continuing to look for a much more rapid delivery of new products from test into 'live', test automation is becoming an increasingly important aspect of the work.

Allied to this is the need to continually assess the impact of other programme roll-out schedules (e.g. CSS) and other BT units (e.g. FBR) on the current testing plans. A slip in the delivery of just one component can cause a completely different set of compatibilities to be verified and/or billing integrity to be checked.

Early involvement of the operational units with the testing process, whether this be with a view to platform management (e.g. ASG, CSNOU [9]) or operation (e.g. Operator Services), has paid dividends for both parties. It has reduced test environment configuration times and smoothed the hand-off from integration to operations as well as allowing more representative training.

In summary, the existing CSP integration team continues to make efficiency improvements almost every month, but the overall approach is now clearly set in a mould that delivers a quality system to an acceptable time frame.

As a final word, the delivery of so large a system through a formal developing strategy has proved the strength of established techniques. Management of the ongoing scope of new releases is key to continuing future success.

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